

Zomato Restaurant Data Analysis

1. Objective

The objective of this project is to analyze restaurant and customer review data from Zomato to identify patterns in restaurant ratings, cuisine preferences, location hotspots, and pricing trends. The analysis aims to generate actionable business insights and recommendations for improving customer engagement and restaurant discovery.

2. Dataset Overview

Dataset Source

Zomato Restaurant Dataset (Kaggle)

Dataset Description

The dataset contains information about restaurants including:

- Restaurant Name
- Location
- Cuisine Type
- Ratings
- Approximate Cost for Two
- Online Ordering Availability
- Customer Reviews

Analysis Tools Used

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - WordCloud
 - Google Colab
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3. Data Cleaning

The following preprocessing steps were performed:

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- Removed duplicate records.
- Handled missing values.
- Converted rating values into numerical format.
- Cleaned cost-related columns.
- Standardized categorical data.
- Verified data consistency and integrity.

Dataset Preview

...	address	name	online_order	book_table	rate	votes	phone	location	rest_type	dish_liked	cuisines	approx_cost(for two people)	listed_in(type)
0	942, 21st Main Road, 2nd Stage, Banashankari, ...	Jalsa	Yes	Yes	4.1/5	775	080 42297555/r/n+91 9743772233	Banashankari	Casual Dining	Pasta, Lunch Buffet, Masala Papad, Paneer Laja...	North Indian, Mughlai, Chinese	800	Buffet
1	2nd Floor, 80 Feet Road, Near Big Bazaar, 6th ...	Spice Elephant	Yes	No	4.1/5	787	080 41714161	Banashankari	Casual Dining	Momos, Lunch Buffet, Chocolate Nirvana, Thai G...	Chinese, North Indian, Thai	800	Buffet
2	1112, Next to KIMS Medical College, 17th Cross...	San Churro Cafe	Yes	No	3.8/5	918	+91 9663487993	Banashankari	Cafe, Casual Dining	Churros, Cannelloni, Minestrone Soup, Hot Choc...	Cafe, Mexican, Italian	800	Buffet
3	1st Floor, Annakuteera, 3rd Stage, Banashankar...	Addhuri Udupi Bhojana	No	No	3.7/5	88	+91 9620009302	Banashankari	Quick Bites	Masala Dosa	South Indian, North Indian	300	Buffet
4	10, 3rd Floor, Lakshmi Associates, Gandhi Baza...	Grand Village	No	No	3.8/5	166	+91 8026612447/r/n+91 9901210005	Basavanagudi	Casual Dining	Panipuri, Gol Gappe	North Indian, Rajasthani	600	Buffet

```
df = pd.read_csv("zomato.csv")
```

Missing Values Analysis

```
df.isnull().sum()
```

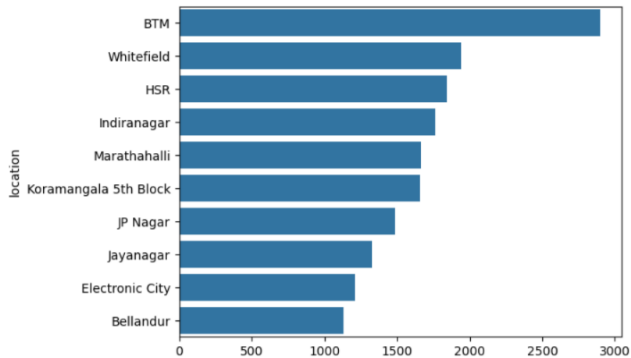
	0
address	17
name	16
online_order	19
book_table	58
rate	7838
votes	78
phone	1296
location	126
rest_type	338
dish_liked	28225
cuisines	203
approx_cost(for two people)	521
listed_in(type)	4610

dtype: int64

4. Visualizations

A. Top Restaurant Locations

```
sns.barplot(x=location_count.values,  
            y=location_count.index,  
            plt.show())
```

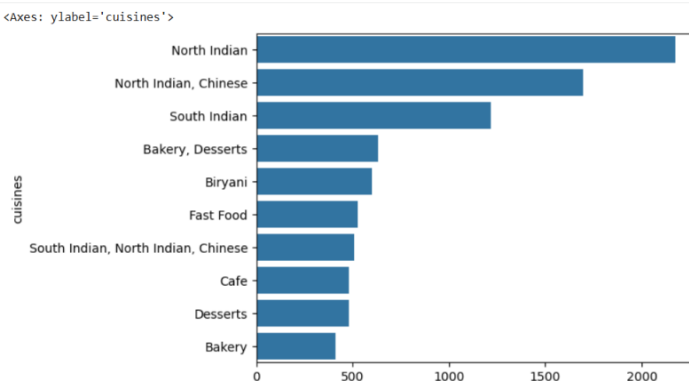


Insight

The analysis identified locations with the highest concentration of restaurants, indicating major food hubs and customer activity zones.

B. Most Popular Cuisines

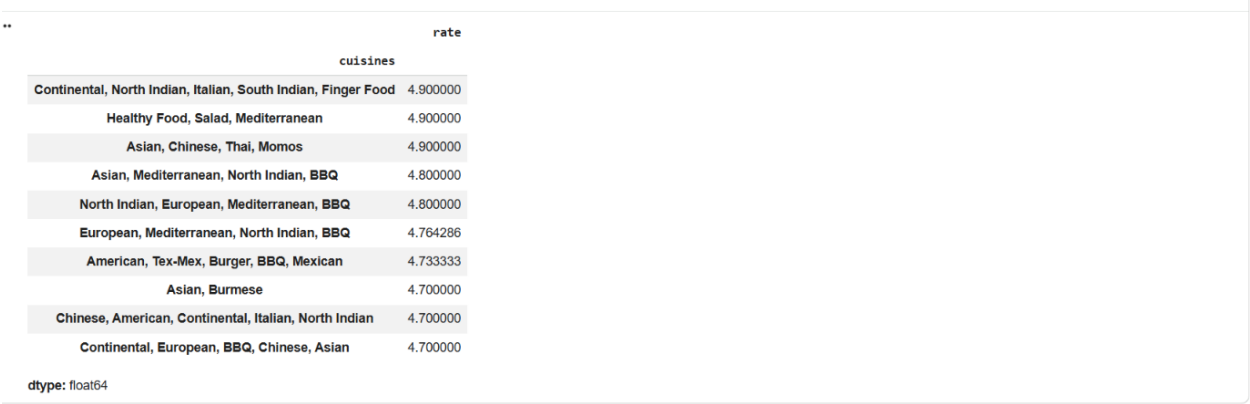
```
cuisine_count = df['cuisines'].value_counts().head(10)  
sns.barplot(x=cuisine_count.values,  
            y=cuisine_count.index)
```



Insight

Certain cuisine categories dominate customer demand and restaurant offerings.

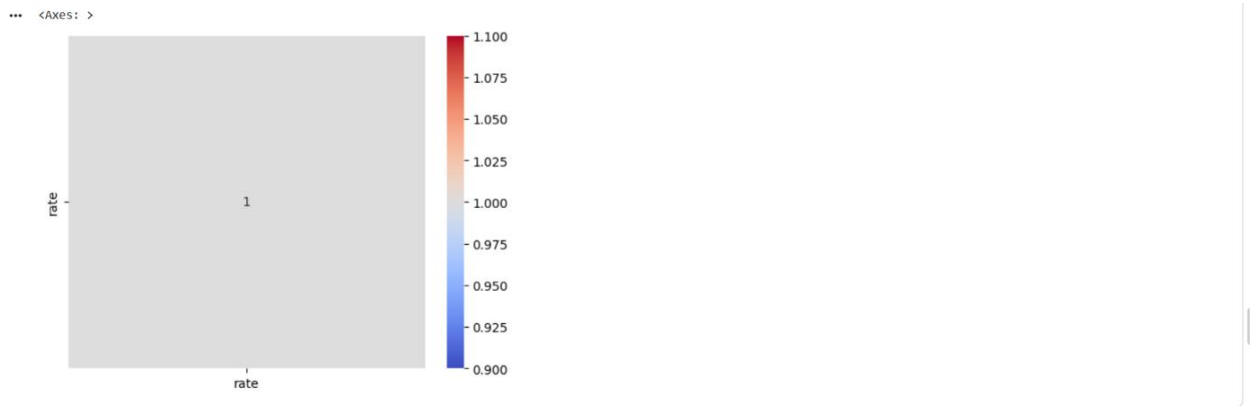
C. Rating Distribution



Insight

Most restaurant ratings fall within the moderate-to-high satisfaction range.

D. Cost vs Rating Analysis



Insight

Higher-priced restaurants do not necessarily achieve better customer ratings.

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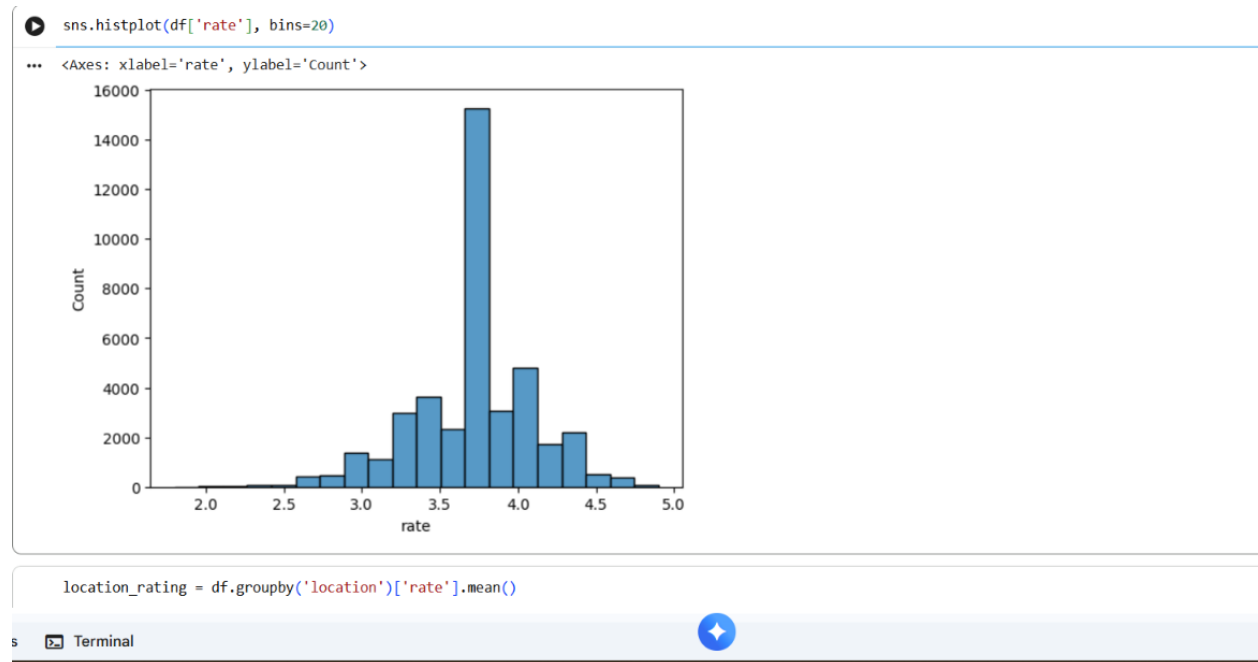
E. Correlation Heatmap



Insight

The heatmap highlights relationships between numerical variables such as cost, votes, and ratings.

F. Cuisine Word Cloud



Insight

Frequently occurring cuisines are visually represented, helping identify customer preferences.

5. Key Findings

1. Restaurant activity is concentrated in specific high-demand locations.
2. North Indian, Chinese, and multi-cuisine restaurants are among the most common categories.
3. Most restaurants maintain ratings between 3.5 and 4.5.
4. Pricing is not the sole factor influencing customer satisfaction.
5. Customer preferences vary across cuisines and locations.
6. Online ordering contributes positively to restaurant visibility and engagement.

6. Business Recommendations

Recommendation 1

Partner with highly rated restaurants to improve platform credibility and customer trust.

Recommendation 2

Promote top-performing cuisines through featured campaigns and personalized recommendations.

Recommendation 3

Implement location-based restaurant recommendation systems for better customer experience.

Recommendation 4

Leverage customer reviews to provide personalized dining suggestions.

Recommendation 5

Collaborate with popular restaurants and influencers to increase platform engagement.

7. Conclusion

This project successfully analyzed restaurant data to identify customer preferences, cuisine trends, rating patterns, and location hotspots. Through data cleaning, visualization, and exploratory analysis, valuable business insights were generated that can support strategic decision-making, improve customer satisfaction, and enhance restaurant discovery systems.

The findings demonstrate the importance of data-driven decision-making in the food delivery and restaurant industry.